

fse supplementary

October 2026

1 Ambiguity Taxonomy

Table 1 presents the ambiguity taxonomy used to annotate the requirements and includes the definition and an example seen in our annotation for each ambiguity type.

2 Wrong Reasoning Taxonomy

Table 2 presents the taxonomy used to annotate the reasoning behind the LLMs' incorrect evaluation and includes the definition and an example for each observed reasoning pattern.

Table 1: Ambiguity taxonomy used to annotate the requirements. The six categories, their definitions, and illustrative examples are adapted from the framework presented in 8 [?]; the examples are instantiated on the requirements analyzed in this paper.

Type	Definition	Example
Lexical	A word used to describe something in the requirement could, in the given context, carry more than one meaning.	<i>“Program”—a software (code) program versus a program of study.</i>
Syntactic	The sentence forks, or a word/adjective applies to more than one other part of the sentence, admitting more than one grammatical parse.	<i>“... monitoring log-in attempts and reporting discrepancies.”</i>
Semantic	The sentence exhibits neither lexical nor syntactic ambiguity, yet still admits multiple interpretations—typically because resolving it requires further context found outside the sentence.	<i>“The system shall implement a mechanism to encrypt and decrypt electronic protected health information (ePHI).”</i>
Vagueness	The sentence uses an adjective that is not wrong but is under-specified: it should give more precise detail so as to remove borderline cases or relative interpretation.	<i>“The system shall terminate a user session after an appropriate period of inactivity”—“appropriate” is itself fuzzily defined.</i>
Incompleteness	An action noun is present (e.g., an emergency) but the requirement does not specify <i>when</i> , during that action, it must be fulfilled.	<i>“The system shall terminate a user session after a predetermined period of inactivity”—the period is never stated.</i>
Referential	Incorrect grammar, or an otherwise unclear antecedent, makes it ambiguous what a noun or pronoun refers to.	<i>“The system shall allow a personal representative the same level of access as a person over their PHI.”</i>

Table 2: FN rejection-reason taxonomy used to annotate the LLM’s incorrect requirement-conformance judgments. The categories and definitions are adapted from the framework of [?]. Examples are taken on the requirements analyzed in this paper. Categories not observed in our annotations are left unfilled.

Type	Definition	Example
Misread Spec	The model misreads what the requirement asks, or points to something that satisfies a different requirement.	Citing “ <i>FORM auth. . . salted SHA-256. . . login timeout</i> ” for a requirement about corroborating that ePHI has not been altered—authentication, not integrity.
Added Requirement	The model adds controls, or hallucinates ones the requirement is not about, inventing capabilities the requirement never mentions.	Rejecting access limited to authorized users and programs for having “ <i>no. . . OAuth/mTLS</i> ” machine-client authentication the requirement never asks for.
Overthink Edge	The model fixates on more detail or a rare edge of something the requirement does cover; the detail can belong to the requirement, but the ground truth shows it is fine to skip, yet the model still rejects for its absence.	Passwords are salted and SHA-256 hashed (accepted by the ground truth), yet rejected for “ <i>no key-stretching/PBKDF.</i> ”
Assumed Type	-	No example found.
Imagined Runtime	-	No example found.
Performance Nitpick	-	No example found.
Readability Nitpick	-	No example found.
Precision Error	-	No example found.
Boundary Error	-	No example found.
Logic Error	The model is wrong about the code: it misses something that is actually there or misstates a capability.	Claiming “ <i>no. . . completeness. . . validation found</i> ” when a server-side validator does run.
Vague Description	The model rejects on a vague concern with no concrete example.	Downgrading for “ <i>some legal phrasing</i> ” without naming any unclear wording.
Other	-	No example found.